

Chapter 3.6 – Modeling in Machine Learning

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Model Selection • Model Evaluation • Re-sampling Methods

Learning Objectives

Explain why model selection and model evaluation are critical after model training.

Distinguish model performance from model complexity.

Explain training error, test error and Mean Squared Error (MSE).

Describe re-sampling as a model-selection strategy.

Apply Holdout, K-fold, Stratified K-fold and LOOCV.

Interpret classifier evaluation using a contingency table, ROC/AUC and precision–recall.

Understand scoring approaches such as Minimum Description Length (MDL).

3.6 Modeling in Machine Learning

Modeling means training a machine-learning algorithm using a training dataset, tuning it to improve performance, validating it, and making predictions on unseen data.

The book summarizes four basic steps: choose a suitable algorithm; train it; tune the model parameters; evaluate the learned model.

Model selection and evaluation are separate but connected activities: the selected model must also demonstrate useful generalization.



Model Selection: What Does It Mean?

Model selection is the process of selecting one good-enough model from alternative machine-learning models for a dataset.

It can also mean selecting features or hyperparameters for the same learning algorithm.

The book emphasizes that finding an absolutely best model is difficult because predictive error is present in practical learning problems.

Goal: choose a model that performs fairly well and generalizes to unseen data.

Two Types of Complexity to Consider

Model Performance

How well does the model perform on the training dataset?

Performance is observed through measures such as error or accuracy.

Good training performance alone is not sufficient.

Model Complexity

How complex is the learned model?

Complex models can fit training data closely.

Excessive complexity may reduce generalization and lead to overfitting.

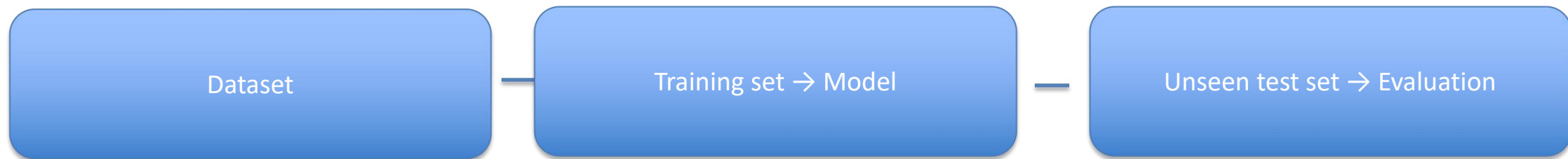
Training Set and Test Set

Training dataset: used to learn the model.

Test dataset: kept unseen during training and used to assess the model's ability to predict new data.

If training and test data are treated as the same data, the model can overfit and may perform poorly on genuinely unseen observations.

Separation of data supports an unbiased estimate of generalization performance.



Training Error, Test Error and Loss

Training error (in-sample error): error obtained when the learned model is applied to the training data.

Test error (out-of-sample error): average error when the model predicts unseen observations.

Loss/error function aggregates differences between true and predicted outputs.

For regression, the book introduces Mean Squared Error (MSE).

Smaller MSE means lower squared prediction error and therefore a more accurate prediction under this criterion.

Average squared difference between observed and predicted values.

$$\text{MSE} = (1/N) \sum [y_i - f(x_i)]^2$$

Model Selection Approaches in the Book

1. Re-sampling methods: construct training/testing/validation sets and observe model performance across the phases.
2. Fit a model on training data and compute measures such as error or accuracy.
3. Use a probabilistic framework and quantify model performance as a score.

The first approach is especially useful when the dataset is relatively small.

3.6.2 Re-sampling Methods

Re-sampling selects or reconstructs training/test data by repeatedly choosing instances from the given dataset.

Purpose: tune/select a model and improve the reliability of estimated performance.

Common methods named in the book: random train/test splits, cross-validation and bootstrap.

Cross-validation family discussed here: Holdout, K-fold, Stratified K-fold and LOOCV.

Holdout Method

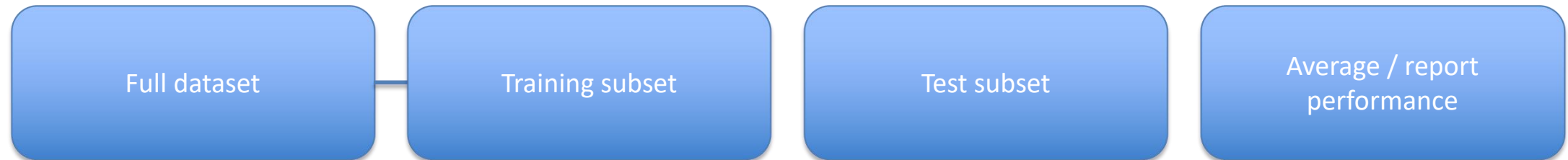
The dataset is split into two subsets: training dataset and test dataset.

The model is trained on the training subset and evaluated on the test subset.

Single holdout: the split is performed once.

Repeated holdout: the split/evaluation is repeated and average performance is used.

Limitation emphasized by the book: performance can have high variance because it depends strongly on the particular split.



K-fold Cross-Validation

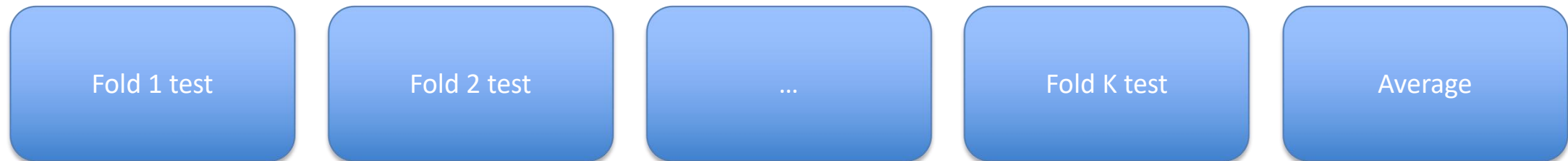
Split the training dataset into K approximately equal folds.

At each iteration, $K-1$ folds are used for training and the remaining fold is used for evaluation.

Repeat for K iterations so that each fold serves as the hold-out fold once.

Average the K performance values to obtain the final estimate.

Advantage: uses the available training data more efficiently than a single holdout.



K-fold Cross-Validation – Algorithm

Input: dataset with n instances and a chosen number K of folds.

1. Split the dataset into K folds.
2. For each fold: train on the other $K-1$ folds.
3. Evaluate on the remaining hold-out fold.
4. Record the performance measure.
5. Compute the average performance across the K iterations.

Stratified K-fold Cross-Validation

Stratified K-fold is similar to ordinary K-fold cross-validation.

Key difference: each fold is constructed to preserve approximately the same proportion of instances for a given categorical/class value.

This is especially useful for classification datasets where class proportions should be maintained across folds.

Example: if the complete dataset has two classes with an imbalanced proportion, each fold attempts to preserve that proportion.

Leave-One-Out Cross-Validation (LOOCV)

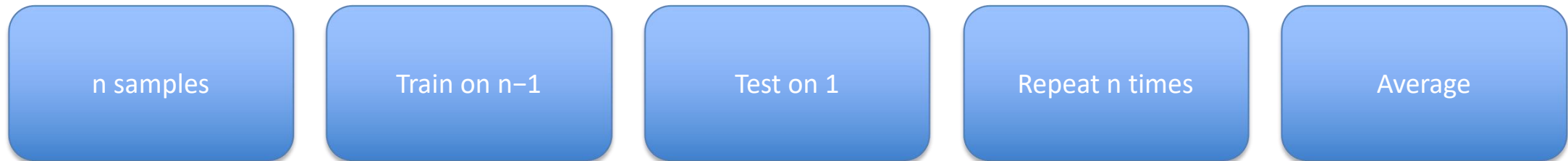
With n instances, LOOCV trains on $n-1$ instances and leaves one instance as the hold-out evaluation case.

The process is repeated n times, with a different instance held out each time.

The average test error/performance across all n runs is the estimate.

Book example: with 100 instances, 99 are used for training and one for testing in each of 100 iterations.

LOOCV is computationally expensive but has low bias.



Worked Example: 5-fold CV

Suppose five folds produce accuracy values: 0.82, 0.86, 0.84, 0.88 and 0.80.

Average accuracy = $(0.82 + 0.86 + 0.84 + 0.88 + 0.80) / 5$.

Average accuracy = $0.84 = 84\%$.

Interpretation: the cross-validation estimate summarizes performance across all five hold-out folds rather than relying on one split.

Use the same metric consistently across all folds.

$$\text{CV Accuracy} = (\sum \text{Accuracy}_i) / K = 0.84$$

Classifier Evaluation: Contingency Table

TP (True Positive): positive cases correctly classified.

TN (True Negative): negative cases correctly classified.

FP (False Positive): negative case incorrectly classified as positive.

FN (False Negative): positive case incorrectly classified as negative.

The book uses disease/cancer testing as an illustrative scenario and notes that FP and FN can have different practical costs.

Actual Positive

Predicted Positive → TP

Predicted Negative → FN

Actual Negative

Predicted Positive → FP

Predicted Negative → TN

Sensitivity and Specificity

Sensitivity is also called the True Positive Rate.

It measures the ability to identify positive cases correctly.

Specificity measures the ability to identify negative cases correctly.

These metrics should be interpreted together when both types of error matter.

$$\text{Specificity} = TN / (TN + FP)$$

$$\text{Sensitivity} = TP / (TP + FN)$$

Precision, Recall and F1 Score

Precision is the positive predictive value: among predicted-positive cases, how many are actually positive?

Recall is the same as sensitivity.

F1 score combines precision and recall through their harmonic mean.

F1 is useful when a balance between precision and recall is needed for a chosen threshold.

$$F1 = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

$$\text{Precision} = TP / (TP + FP) \quad | \quad \text{Recall} = TP / (TP + FN)$$

ROC Curve and AUC

ROC is a visual method for checking and comparing classifier performance.

It plots sensitivity (TPR) against false positive rate ($1 - \text{specificity}$) for different threshold values.

The book illustrates several classifiers and notes that a larger area under the ROC curve indicates better performance.

AUC = area under the ROC curve; a perfect classifier has $\text{AUC} = 1$.

Threshold changes

TPR & FPR

ROC curve

Area under curve $\rightarrow$ AUC

Precision–Recall Curve

A precision–recall curve plots precision and recall for different classification thresholds.

It is particularly useful when class imbalance is present.

The book contrasts ROC and precision–recall usage: ROC is used when there is no class imbalance, while precision–recall is useful for moderate-to-large class imbalance.

Threshold selection controls the trade-off between false positives and false negatives.

Scoring Methods: Minimum Description Length

Model selection can combine model complexity and performance into a single score.

Minimum Description Length (MDL) describes the target variable and model using bits.

MDL follows the idea that a simpler explanation/model is preferable when it provides an adequate description of the data.

It is presented as an alternative scoring-based approach for model selection.

Comparison of Re-sampling Methods

Holdout / Repeated Holdout

Simple and fast

Performance depends on split

Repeated holdout reduces dependence on one split

K-fold / Stratified / LOOCV

K-fold: K training/evaluation cycles

Stratified: preserves class proportions

LOOCV: n cycles; low bias but expensive

Exam-oriented Summary – Section 3.6

Define model selection and model evaluation.

Differentiate training error and test error.

Derive/compute MSE for a set of predicted and actual values.

Explain Holdout, K-fold, Stratified K-fold and LOOCV.

Construct a contingency table and compute sensitivity, specificity, precision, recall and F1.

Explain ROC, AUC and precision–recall curves.

Discuss MDL as a scoring approach.